

Power BI Assignment 1 – Data Transformation & Data Modeling

E-Commerce Sales Analysis

This assignment will help you explore e-commerce sales data analysis using Power BI. Below are the files you will be working with (click on each to download):

- ★ [List of Orders.csv](#)
- ★ [Order Details.csv](#)
- ★ [Sales target.csv](#)

In this exercise, you will leverage Power BI's capabilities to import, transform, model, and analyze the provided data.

Instructions:-

Import Data:

- Import “List of Orders.csv” into Power BI.
 - GetData->import the file
- Open “List of Orders” in Power Query Editor by clicking on ‘Transform’.
- Import “Order Details.csv” and “Sales target.csv” into Power Query Editor.

Data Transformation:

- Restrict the "List of Orders" table to only the first 500 rows.
Home>Keep rows->Keep Top Rows->500
- Ensure the “Order Date” column in the “List of Orders” table is set to data type 'Date'.
Orderdate Colm->right click->Change type->Date(can be seen on left side of colm header)
- Change the data type of “Amount” and “Target” columns to ‘Fixed Decimal Number’.
Right click the columns → Change data type -> Fixed Decimal Number

- Format the "CustomerName" column into proper case, ensuring consistent capitalization for each word.

Transform → Format → Capitalize each word

- Merge the "State" and "City" columns to create a new column named "Location" in the format 'City, State'.

Select columns → Add Column → Merge column → Select separator and add new column name

- Create a new custom column named "Profit Margin" as the percentage of "Profit" divided by "Amount".

Add column → Custom column → Give custom formula and click ok → Changed data type to percentage

- Add a new conditional column named "Profit Status" based on the values in the "Profit" column. The conditions are as follows: if the profit is less than 0, the label should be "Loss"; if the profit equals 0, the label should be "Break-Even"; and if the profit is greater than 0, the label should be "Profit".

Add column → Conditional column → Give conditions → Ok

Merging Data (Joins):

- Merge the "List of Orders" and "Order Details" tables into a new single table named "Orders Data" based on the "Order ID" relationship.

Home → Merge query → Merge query as new → Select tables and highlight the common column → Join kind(Left outer) → Ok

On new table expand the new table → Tick the required columns-> Untick use original column name as prefix

Change Table name

Handling Missing Data & Duplicate Data:

- Identify missing values in the data and determine a strategy to address them.

Click o the column header filter →Find if any null or blank values

Right click →Replace values →Blank with 0 for numerical and unknown for text

- Check for duplicate rows and define a strategy to handle duplicates.

Select all the columns →Remove rows →Remove duplicate rows

Sorting and Filtering Data:

- In the ‘Orders Data’ table, utilize sorting and filtering techniques on columns like Order Date, State or Category to analyze data based on specific criteria:

- ◆ Sort the orders by Order Date in descending order to analyze recent trends.

Select column →Column header dropdown →Sort descending

- ◆ Filter the orders to focus only on a specific state (e.g., Tamil Nadu) for regional analysis.

Select column →Column header dropdown →Tick the value, needed to be filtered

Grouping and Aggregating Data:

- Duplicate the “Order Details” table and calculate the count of each Order ID, average profit by Category or total amount by Sub-Category.

Right click tbl_name →duplicate →Home →Transform →Group by

In basic, give the required column name, functionality and new column name

- Duplicate the “Sales Target” table and aggregate the total target amount by Month of Order Date.

Right click tbl_name →duplicate →Home →Transform →Group by

In basic, give the required column name, functionality and new column name

CLOSE AND APPLY

Data Modeling:

- Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.

In power bi desktop, select model view

Select column name from required table and drag it to the same column name in the connection table

- Build a relationship between the “Order Details” and “Sales Target” tables based on the ‘Category’ column. Click "Manage relationships" and ensure this relationship is active.

In power BI desktop, select model view,

Select column name from required table and drag it to the same column name in the connection table

In Home →Manage relationships->Ensure all relationships are active and cardinality is correct.